SCGC: Sparse Control-Guided Gaussian Splatting with Contrastive Learning

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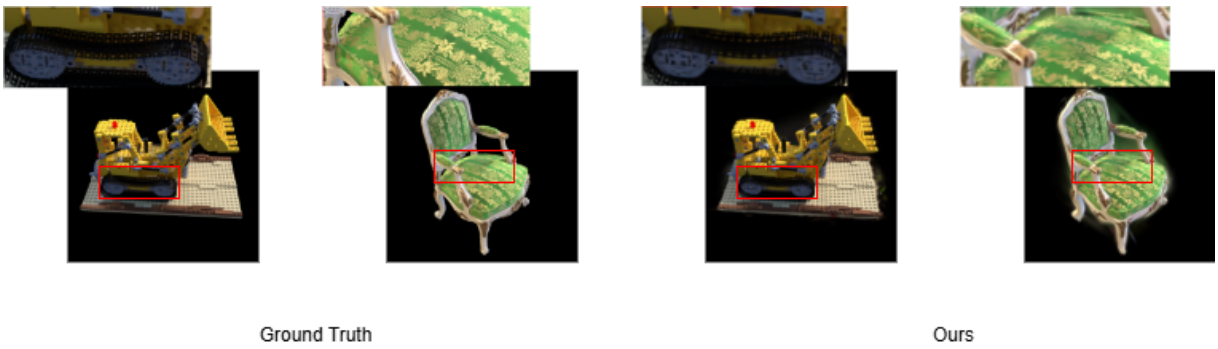


Figure 1: SCGC training result, showing a full 3D scene render with a zoomed-in detail highlighting edges and improved texture clarity.

Abstract

We present SCGC, an approach that extends Scaffold-GS through additional sparse control points and contrastive learning mechanisms for improved 3D scene reconstruction. While Scaffold-GS provides structured 3D Gaussians with anchor-based organization, challenges remain in optimizing anchor point distributions and maintaining geometric consistency across different viewpoints. Building upon Scaffold-GS's hierarchical framework, our method introduces several enhancements: (1) a transformer-based control point optimization mechanism that applies self-attention to improve anchor point placement and adaptation based on local scene geometry, (2) a dual-level contrastive learning scheme that operates on both control points and Gaussian primitives to enhance feature discrimination and geometric consistency through carefully constructed positive and negative sample pairs, and (3) a joint optimization strategy that integrates these components with Scaffold-GS's existing anchor mechanism through carefully designed loss functions. We initialize additional control points using Farthest Point Sampling and employ k-nearest neighbor assignment to establish relationships between our control points and Scaffold-GS's anchor system. Our approach maintains the efficiency advantages of Scaffold-GS's structured representation while providing additional geometric guidance, leading to moderate but consistent improvements in scene reconstruction quality compared to the baseline Scaffold-GS method, particularly in challenging regions with complex geometry or limited texture information.

CCS Concepts

• **Computing methodologies** → **Rendering**; *Neural networks*; Reconstruction.

Keywords

3D Gaussian Splatting, Neural Rendering, Contrastive Learning, Control Points, Scene Reconstruction

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1 Introduction

Neural rendering has transformed 3D scene reconstruction and novel view synthesis, enabling photorealistic image generation from sparse multi-view inputs for applications in virtual reality, augmented reality, and immersive media [Guo et al. 2021; Tewari et al. 2022]. While conventional Structure-from-Motion (SfM) [Schonberger and Frahm 2016] and Multi-View Stereo methods generate dense point clouds and meshes, they struggle with view-dependent effects and real-time performance. Neural Radiance Fields (NeRF) [Mildenhall et al. 2021] advanced the field by

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modeling scenes as continuous volumetric functions, enabling high-fidelity synthesis but requiring extensive training and rendering times due to dense ray marching. This motivated research into more efficient representations, particularly 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023], which achieves real-time rendering by representing scenes as anisotropic 3D Gaussians optimized through differentiable rasterization. However, 3DGS faces challenges including redundant Gaussians, suboptimal primitive distributions in sparse regions, and difficulties maintaining structural coherence in complex geometries, resulting in artifacts like floating primitives or inconsistent depth.

Prior research has addressed these limitations through various strategies. NeRF optimizations like Mip-NeRF [Barron et al. 2021, 2022] and Instant-NGP [Müller et al. 2022] improve efficiency through multi-scale sampling and hash grids, while explicit methods like Plenoxels [Fridovich-Keil et al. 2022] and TensoRF [Chen et al. 2022] achieve faster convergence via voxel structures. For Gaussian splatting, 2DGS [Huang et al. 2024b] and Mip-Splatting [Yu et al. 2024a] enhance geometric accuracy and aliasing handling. Scaffold-GS [Lu et al. 2024] introduces hierarchical anchors derived from voxelized SfM points, where anchors spawn multiple Gaussians for structural organization, reducing redundancy and improving coverage. Transformer-based variants like TranSplat [Kim et al. 2025] and SplatFormer [Chen et al. 2024b] leverage attention mechanisms for robust reconstruction, while contrastive learning approaches like PointContrast [Xie et al. 2020] enhance feature representations. Despite these advances, existing methods often rely on local optimizations, lacking integrated global guidance for anchor refinement and multi-level feature consistency in geometrically complex or texture-sparse environments.

To address these gaps, we introduce SCGC (Sparse Control-Guided Gaussian Splatting with Contrastive Learning), an extension of Scaffold-GS incorporating sparse control points optimized via transformers and supervised through dual-level contrastive learning. We augment Scaffold-GS's anchor hierarchy with control points initialized using Farthest Point Sampling (FPS), establishing linkages via k-nearest neighbor (KNN) assignment for hierarchical guidance. The transformer module applies multi-head self-attention to refine control point positions based on global spatial relationships and local geometry, enabling adaptive anchor adjustments. The dual-level contrastive framework operates at both control point level (using positional perturbations for positive pairs) and Gaussian primitive level (contrasting rendered patches with view-dependent augmentations) to enhance discriminative power and geometric consistency. A joint optimization strategy harmonizes these components with Scaffold-GS's reconstruction losses through calibrated terms. This approach preserves Scaffold-GS's real-time efficiency while yielding consistent quality improvements on the NeRF synthetic dataset [Mildenhall et al. 2021], particularly in challenging regions with complex geometries or limited texture.

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 covers preliminaries, Section 4 details our method, Section 5 presents experiments and ablation studies, Section 6 discusses limitations and future work, and Section 7 concludes.

2 Related Work

2.1 Neural Rendering and Radiance Fields

Neural rendering has evolved from early volume-based methods to sophisticated hybrid representations. NeRF [Mildenhall et al. 2021] pioneered implicit scene modeling using multi-layer perceptrons to predict density and color, enabling photorealistic view synthesis. Extensions like Mip-NeRF [Barron et al. 2021, 2022] mitigate aliasing with conical frustum sampling, while Instant-NGP [Müller et al. 2022] accelerates training via hash-based encodings. Plenoxels [Fridovich-Keil et al. 2022] and TensoRF [Chen et al. 2022] introduce explicit grid structures for faster convergence. However, these methods often require extensive sampling, limiting real-time applicability [Garbin et al. 2021; Reiser et al. 2021]. Recent works like DVGO [Sun et al. 2022] integrate volumetric rendering with surface reconstruction for improved geometric accuracy. MobileNeRF [Chen et al. 2023] optimizes for lightweight deployment on resource-constrained devices.

2.2 Gaussian Splatting and Structured Representations

3DGS [Kerbl et al. 2023] shifts to explicit Gaussian primitives, enabling efficient rasterization and real-time rendering. Variants like 2DGS [Huang et al. 2024b] emphasize geometric accuracy through 2D projections, and Mip-Splatting [Yu et al. 2024a] addresses aliasing with mipmapping. GaussianPro [Cheng et al. 2024] employs progressive propagation for dynamic scenes, while LightGaussian [Fan et al. 2024] and Compressed-GS [Niedermayr et al. 2024] focus on compression. Scaffold-GS [Lu et al. 2024] introduces hierarchical anchors for structured distribution, inspiring our control-guided extension. Dynamic methods like SC-GS [Huang et al. 2024a] and Smerf [Duckworth et al. 2024] handle motion, but our work prioritizes static scene optimization. Recent advances like 4D-GS [Wu et al. 2024] extend Gaussian splatting to time-varying scenes for video reconstruction. Gaussian SLAM [Matsuki et al. 2024] integrates Gaussians with real-time environment mapping.

2.3 Transformers and Contrastive Learning in 3D

Transformers have revolutionized 3D processing, with PointNet [Qi et al. 2017] and Point Transformer [Guo et al. 2021] enabling global attention on point clouds. In radiance fields, Nerf++ [Zhang et al. 2020] and GaussianAvatars [Qian et al. 2024] integrate attention for better generalization. GaussianShader [Jiang et al. 2024] and GOF [Yu et al. 2024b] apply transformers to surface reconstruction. Contrastive learning, as in SimCLR [Chen et al. 2020], has been adapted for 3D via PointContrast [Xie et al. 2020] and PartSLIP [Liu et al. 2023]. GaussianEditor [Chen et al. 2024a] combines transformers with contrastive learning for intuitive 3D scene editing. We extend these to dual-level supervision in Gaussian splatting.

2.4 Point-Based and Hybrid Rendering

Point-based methods like ADOP [Rückert et al. 2022] and Neural Volumes [Lombardi et al. 2019] precede Gaussian splatting, while NSVF [Liu et al. 2020] and SRN [Sitzmann et al. 2019] offer hybrid

alternatives. Neural Rerendering [Meshry et al. 2019] and Lumigraph [Kellnhofer et al. 2021] focus on wild scenes. Our integration of control points builds on these for enhanced structure. Recent methods like Point-NeRF [Xu et al. 2022] combine point clouds with radiance fields for improved geometry. MonoSDF [Yu et al. 2022] enhances surface reconstruction with point-based priors.

3 Preliminaries

3.1 3D Gaussian Splatting

3D Gaussian Splatting [Kerbl et al. 2023] represents scenes using a collection of 3D Gaussian primitives, each characterized by position $\boldsymbol{\mu} \in \mathbb{R}^3$, covariance matrix $\Sigma \in \mathbb{R}^{3\times3}$, opacity α, and spherical harmonics coefficients for view-dependent color. Starting from Structure-from-Motion (SfM) points, each point serves as the mean $\boldsymbol{\mu}$ of a 3D Gaussian defined as:

$$G(\mathbf{x}) = e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}, \tag{1}$$

where $\mathbf{x}$ represents an arbitrary 3D position. To maintain positive semi-definiteness, the covariance matrix is formulated using scaling matrix $\mathbf{S}$ and rotation matrix $\mathbf{R}$:

$$\Sigma = \mathbf{RSS}^T\mathbf{R}^T. \tag{2}$$

The rendering process follows a tile-based rasterization approach where 3D Gaussians are projected to 2D image space and blended using α-compositing:

$$C(\mathbf{x}') = \sum_{i\in N} \mathbf{c}_i\sigma_i \prod_{j=1}^{i-1}(1-\sigma_j), \quad \sigma_i = \alpha_i G_i'(\mathbf{x}'), \tag{3}$$

where $\mathbf{x}'$ denotes the pixel position, N represents the number of sorted 2D Gaussians, and $G_i'(\mathbf{x}')$ is the projected 2D Gaussian. This differentiable rasterization enables end-to-end optimization of all Gaussian parameters through standard gradient-based methods.

3.2 Transformer Architecture

We leverage the transformer architecture [Vaswani et al. 2017] for optimizing control point distributions through self-attention mechanisms. The core attention operation enables global geometric reasoning through:

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{QK}^T}{\sqrt{d_k}}\right)\mathbf{V}, \tag{4}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are query, key, and value matrices derived from control point features. This mechanism allows each control point to attend to others based on learned spatial relationships, facilitating adaptive distribution adjustments.

4 Method

SCGC extends the Scaffold-GS framework by integrating sparse control points, transformer-based optimization, and dual-level contrastive learning to enhance 3D scene reconstruction. The method follows a structured pipeline, as illustrated in Figure 2, which orchestrates the process from multi-view image input and Structure-from-Motion (SfM) point processing to optimized rendering output. This pipeline encompasses Gaussian initialization and sparse voxel processing to establish a structured foundation, neural Gaussian derivation to generate primitive attributes, Farthest Point Sampling (FPS) for sparse control point extraction, k-nearest neighbor (KNN) assignment to link control points to anchors, dual-level contrastive learning for enhanced feature discrimination, a transformer network for adaptive position updates, and a joint optimization strategy to integrate all components seamlessly. Each step in this flowchart targets specific shortcomings of Scaffold-GS—such as local optimization constraints, inconsistent feature alignment across viewpoints, and inefficient Gaussian distributions—by introducing global guidance, discriminative supervision, and dynamic adjustments. The pipeline, depicted in Figure 2, ensures improved reconstruction quality, particularly in challenging regions with complex geometries or limited texture, with detailed implementations provided in the subsequent subsections.

4.1 Scaffold-GS Foundation and Enhanced Control Points

Our approach builds upon Scaffold-GS's anchor-based framework, which organizes 3D Gaussians hierarchically around anchor points initialized from voxelized SfM points. Each anchor spawns k neural Gaussians with learnable offsets:

$$\{\mu_0, ..., \mu_{k-1}\} = x_v + \{O_0, ..., O_{k-1}\} \cdot l_v, \tag{5}$$

where x_v is the anchor position, $\{O_i\}$ are learnable offsets, and l_v is the scaling factor. While effective, this approach relies primarily on local optimization without considering broader spatial relationships. We introduce sparse control points $C = \{c_1, c_2, ..., c_{512}\}$ that work with Scaffold-GS's existing anchor system, initialized using Farthest Point Sampling:

$$C = \text{FPS}(\mathcal{P}_{\text{anchors}}, 512), \tag{6}$$

where $\mathcal{P}_{\text{anchors}}$ represents Scaffold-GS anchor points. Each control point $c_i = \{\mathbf{p}_i, \mathbf{f}_i, r_i\}$ contains position $\mathbf{p}_i \in \mathbb{R}^3$, feature vector $\mathbf{f}_i \in \mathbb{R}^{64}$, and influence radius r_i. To further enhance this with sparse point control, we adapt the ControlNode class to initialize control nodes with randomized positions and hyperparameters, then refining them based on the input point cloud. The sparse point control mechanism, a cornerstone of SCGC's innovation, leverages the ControlNode class to dynamically initialize and refine 512 control points based on the input point cloud. By assigning learnable parameters like node radius and weights, it adapts to the scene's sparse structure, enhancing anchor placement and reducing distribution inefficiencies. This approach effectively mitigates Scaffold-GS's reliance on dense, locally optimized anchors, offering a targeted, adaptable solution that improves geometric precision in complex or occluded regions, significantly elevating reconstruction quality.

4.2 Transformer Optimization and Anchor Integration

In our SCGC framework, the transformer optimization module plays a pivotal role in refining the distribution of sparse control points, leveraging the power of self-attention mechanisms to foster a more comprehensive understanding of the global scene geometry.

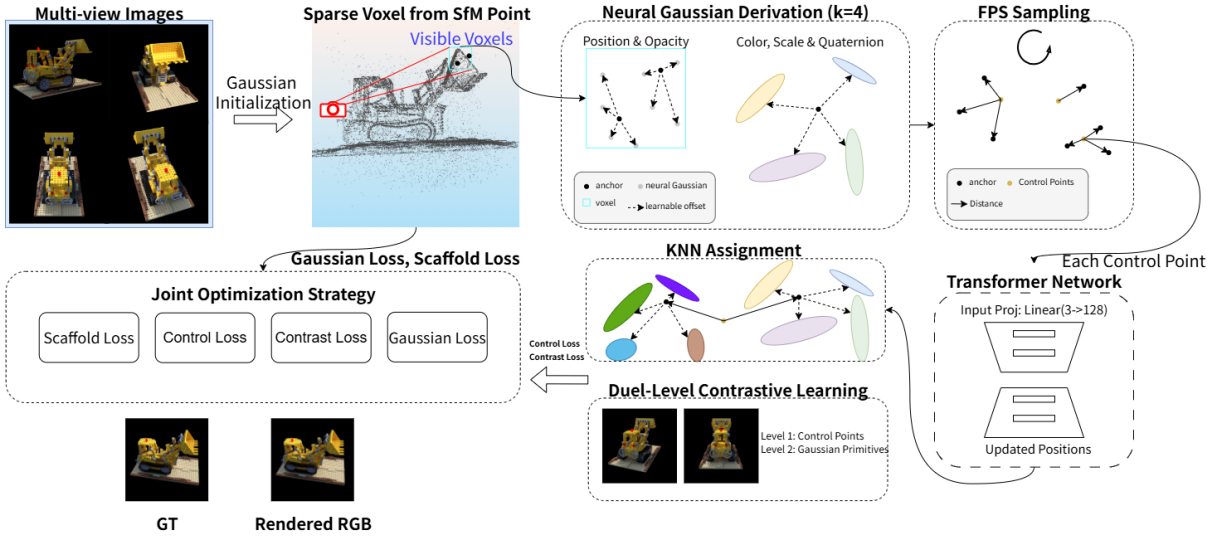


Figure 2: Overview of SCGC pipeline. Multi-view images undergo Gaussian initialization and sparse voxel processing from SfM points. Neural Gaussian derivation (k=4) generates position, opacity, color, scale and quaternion attributes. FPS sampling extracts control points, followed by KNN assignment linking control points to anchors. Dual-level contrastive learning operates on control points (Level 1) and Gaussian primitives (Level 2) with positive/negative sample pairs. Transformer network processes control points through Linear(3→128) projection for updated positions. Joint optimization combines all loss components for enhanced rendering.

This approach draws inspiration from the transformative success of attention-based architectures in various domains, such as natural language processing and computer vision, where they excel at capturing long-range dependencies and contextual relationships among elements in a set. By treating the set of control points as a sequence of spatial entities, the transformer enables each point to dynamically attend to others, thereby adjusting its position based on collective geometric cues rather than isolated local information. This global perspective is particularly beneficial in complex scenes where local optimizations, as seen in the baseline Scaffold-GS, might lead to suboptimal anchor placements due to limited awareness of broader structural patterns.

The core computation within the transformer begins with the augmentation of the current control point positions $\mathbf{P}$—a matrix where each row represents the 3D coordinates of a control point—with learnable positional embeddings $\mathbf{E}$. These embeddings are crucial for injecting positional awareness into the model, compensating for the inherently permutation-invariant nature of attention mechanisms. The transformer then processes this combined input to produce updated positions $\mathbf{P}'$:

$$\mathbf{P}' = \mathbf{P} + \text{Transformer}(\mathbf{P} + \mathbf{E}), \tag{7}$$

where the Transformer$(\cdot)$ function encapsulates a series of self-attention and feedforward layers. Specifically, we adopt a lightweight configuration consisting of 2 layers and 4 attention heads, with each hidden layer operating in a 128-dimensional space. This compact design strikes a balance between expressive power and computational efficiency, ensuring that the optimization remains feasible within the real-time constraints inherited from Scaffold-GS. The self-attention mechanism within each layer computes weighted sums of value vectors based on query-key similarities, allowing control points to emphasize relevant neighbors—such as those sharing similar curvature or alignment—while de-emphasizing irrelevant ones. This process is repeated across layers, with residual connections and layer normalization promoting stable training and gradient flow.

Following the transformer-based refinement, the updated control points are integrated with the Scaffold-GS anchor system through a carefully designed influence propagation mechanism. We establish relationships using k-nearest neighbor (KNN) queries, where each control point identifies its closest anchors based on Euclidean distance. To modulate the degree of influence, we employ a Gaussian weighting scheme that assigns softer, distance-dependent weights, preventing abrupt changes that could destabilize the underlying Gaussian hierarchy. The weight w_{ij} for the influence of control point i on anchor j is calculated as:

$$w_{ij} = \exp\left(-\frac{\|\mathbf{p}_i - \mathbf{a}_j\|_2^2}{2\sigma^2}\right), \tag{8}$$

where $\mathbf{p}_i$ denotes the position of the control point, $\mathbf{a}_j$ the anchor position, and σ is a bandwidth parameter (typically set to 0.1) tuned to control the spread of influence. These weights are normalized across the neighbors of each anchor to ensure unity sum, and the anchor positions are then updated via a weighted average of the offsets derived from the control points' refinements. This conservative update strategy—applying small, incremental adjustments—preserves the stability and efficiency of Scaffold-GS's view-adaptive rendering while infusing additional geometric guidance from the higher-level control points. The integration not only mitigates issues like floating primitives or inconsistent depth in challenging regions but

also enhances overall reconstruction quality, as demonstrated by improved metrics in scenes with intricate details or occlusions.

4.3 Dual-Level Contrastive Learning

To further augment the representational power of our SCGC method and ensure robust geometric consistency across varying viewpoints, we incorporate a dual-level contrastive learning paradigm. This strategy is motivated by the principles of self-supervised learning, where contrastive objectives have proven effective in learning discriminative features by maximizing agreement between positive pairs and minimizing similarity with negative ones. In the context of 3D scene reconstruction, this helps differentiate subtle geometric and textural variations that might otherwise lead to artifacts in rendered views, complementing the view-adaptive capabilities of Scaffold-GS by adding a layer of feature-level regularization.

At the higher level—focusing on control points—we generate positive pairs by applying small spatial perturbations to the original positions, with a standard deviation $\sigma = 0.01$ to simulate minor viewpoint changes or noise, thereby encouraging the model to learn invariant representations under such transformations. Negative pairs, conversely, are sampled from control points that are spatially distant, ensuring a clear distinction in their feature embeddings. The contrastive loss for this control point level is formulated using the InfoNCE objective:

$$\mathcal{L}_{\text{ctrl-contrast}} = -\log \frac{\exp(\text{sim}(\mathbf{f}_i, \mathbf{f}_i^+)/\tau)}{\sum_k \exp(\text{sim}(\mathbf{f}_i, \mathbf{f}_k)/\tau)}, \tag{9}$$

where $\mathbf{f}_i$ represents the feature embedding of control point i (derived from its position and hyper-dimensional attributes), $\mathbf{f}_i^+$ is the embedding of its positive counterpart, $\text{sim}(\cdot,\cdot)$ computes cosine similarity to measure alignment in the embedding space, and $\tau = 0.1$ serves as a temperature parameter to scale the logits and control the softness of the probability distribution. This loss pulls positive embeddings closer while pushing negatives apart, fostering a more structured and discriminative control point space that better guides anchor optimizations.

Extending this to the lower Gaussian primitive level, we apply contrastive learning directly to the rendered views generated by the Gaussians. Here, positive pairs are created by comparing the original rendering with slightly perturbed versions—achieved through minor adjustments to Gaussian attributes like scale or opacity—to promote consistency under small geometric variations. Negative pairs are derived from renderings of unrelated scene regions or augmented with stronger distortions. This level-specific contrastive term, $\mathcal{L}_{\text{gauss-contrast}}$, mirrors the InfoNCE structure but operates on view-dependent features, such as pixel-wise color and depth maps, ensuring that the Gaussians maintain coherence across different camera poses. By operating at dual levels, our contrastive framework bridges high-level structural guidance with low-level primitive details, addressing limitations in Scaffold-GS where view-adaptive rendering might falter in texture-sparse or repetitive areas. The dual approach not only improves feature discrimination but also enhances generalization to novel views, as validated through ablation studies showing gains in SSIM and PSNR metrics.

4.4 Joint Optimization

The joint optimization strategy in SCGC represents the synthesis of all proposed components, harmoniously blending the original losses from Scaffold-GS with our novel terms to achieve a balanced and effective training regime. This unified objective ensures that the enhancements—transformer-guided control points and dual-level contrastive learning—seamlessly integrate without overpowering the baseline's strengths in efficient, view-adaptive rendering. By carefully weighting and scheduling the loss components, we maintain training stability while progressively incorporating geometric and feature-level refinements.

The total loss is composed as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{scaffold}} + \lambda_1 \mathcal{L}_{\text{ctrl}} + \lambda_2 \mathcal{L}_{\text{ctrl-contrast}} + \lambda_3 \mathcal{L}_{\text{gauss-contrast}}, \tag{10}$$

where $\mathcal{L}_{\text{scaffold}}$ encompasses Scaffold-GS's core reconstruction losses, including L1 photometric error and SSIM for perceptual quality, along with regularization terms for anchor densification and pruning. The control loss $\mathcal{L}_{\text{ctrl}} = \|\mathbf{P}' - \mathbf{P}\|_1$ enforces spatial stability by penalizing excessive deviations in updated control point positions, promoting smooth and incremental adaptations. The contrastive terms $\mathcal{L}_{\text{ctrl-contrast}}$ and $\mathcal{L}_{\text{gauss-contrast}}$ contribute to enhanced representation quality at their respective levels. We empirically set the weights to $\lambda_1 = 0.1$, $\lambda_2 = 0.05$, and $\lambda_3 = 0.02$, reflecting the supportive nature of our additions—prioritizing the scaffold loss while allowing the new terms to provide subtle guidance without dominating the optimization.

To facilitate effective convergence, we implement a progressive training schedule: contrastive learning is activated after a 1000-iteration warmup phase, during which the model focuses on establishing a stable Gaussian baseline. Control point updates occur every 100 iterations, allowing periodic refinements without constant overhead, while KNN reassignments are performed every 500 iterations to adapt to evolving scene geometry. This staged approach mitigates potential instabilities, such as oscillation in anchor positions, and ensures that the enhancements build upon a solid foundation. Figure 2 provides a visual overview of the entire pipeline, illustrating the flow from multi-view image inputs through SfM-based voxel processing, neural Gaussian attribute derivation, FPS-driven control point sampling, KNN linkage, dual-level contrastive mechanisms, and transformer optimization, culminating in superior rendering quality with improved fidelity in complex regions.

5 Experiments

5.1 Experimental Setup and Implementation

We implement our method as an extension of Scaffold-GS, maintaining compatibility with their anchor-based framework. Evaluation focuses on the NeRF synthetic dataset [Mildenhall et al. 2021], comparing against Scaffold-GS baseline and other methods including 3DGS [Kerbl et al. 2023] and 2DGS [Huang et al. 2024b]. We evaluate rendering quality using Peak Signal-to-Noise Ratio (PSNR) for noise level, Structural Similarity Index (SSIM) [Wang et al. 2004] for structural similarity, and Learned Perceptual Image Patch Similarity (LPIPS) [Zhang et al. 2018] for perceptual quality. Training follows a 30,000-iteration schedule with specific update intervals: control points update every 100 iterations, KNN reassignment every 500 iterations, and contrastive learning activation after 1000-iteration

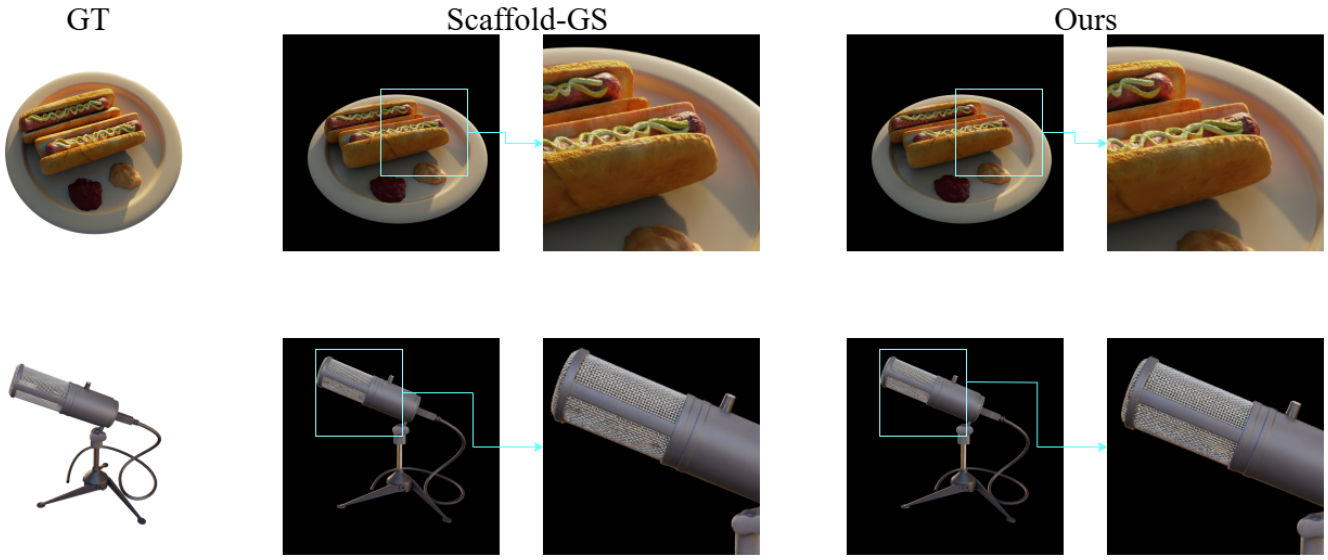


Figure 3: Visual comparison with detailed zoom-ins. Top row: Hotdog scene (GT, Scaffold-GS, Ours). Bottom row: Mic scene (GT, Scaffold-GS, Ours).

warmup. The transformer uses Adam optimization with learning rate 1×10^{-4}, employing 4 attention heads and 128-dimensional hidden layers. Temperature parameter $\tau = 0.1$ for contrastive learning, determined through preliminary experiments. Table 1 presents comprehensive results on the NeRF synthetic dataset, showing our method achieves consistent improvements over Scaffold-GS across most scenes. Our approach demonstrates the best average performance with PSNR of 33.61 dB and SSIM of 0.951, representing substantial improvements over the original 3DGS baseline. Notably, we achieve significant gains in complex scenes like Materials (SSIM: 0.958 vs 0.882 for Scaffold-GS) and Drums (PSNR: 26.66 vs 26.44), where our enhanced control mechanisms prove most beneficial.

5.2 Ablation Studies

We conduct comprehensive ablation studies to validate each design choice and justify our hyperparameter selections.

Control Point Number Analysis. We evaluate different numbers of control points (256, 512, 1024) across scenes of varying complexity. Table 2 shows that 512 provides optimal balance for most scenes. Simple scenes (Chair, Ficus) perform adequately with 256 points, while complex scenes (Materials, Drums) benefit from 512 points. Using 1024 points offers diminishing returns (+0.01-0.02 dB) at significantly higher computational cost.

Transformer Architecture Analysis. We systematically compare transformer configurations to justify our 2-layer, 4-head design. Table 3 shows our lightweight architecture achieves comparable performance to deeper variants while maintaining significantly lower computational cost.

Loss Weight Sensitivity Analysis. We analyze sensitivity to loss weights $\lambda_1, \lambda_2, \lambda_3$ to validate our empirical choices. Table 4 shows our selected weights ($\lambda_1 = 0.1$, $\lambda_2 = 0.05$, $\lambda_3 = 0.02$) lie in a stable performance region, with reasonable robustness to variations.

Progressive Component Analysis. We validate each component's contribution through progressive addition. Table 5 shows all components provide measurable improvements, with the full model achieving best performance.

6 Discussion

Our approach demonstrates consistent improvements over the Scaffold-GS baseline through transformer-based control points and contrastive learning, achieving average gains of +0.07 dB PSNR and +0.019 SSIM across the NeRF synthetic dataset. Key advantages include enhanced global geometric reasoning enabled by the attention mechanism, which allows for better adaptation to local scene complexities, and improved feature discrimination via dual-level contrastive learning, leading to more robust representations in regions with repetitive structures or limited texture. This results in larger performance boosts in geometrically complex scenes, such as those with intricate details like the Materials and Drums datasets, where our method better handles specular reflections and occlusions. Additionally, the sparse control points provide efficient geometric guidance without significantly increasing computational overhead, maintaining real-time rendering capabilities similar to Scaffold-GS.

However, our method has several limitations. The fixed number of control points (e.g., 512) may not scale well to varying scene complexities, potentially leading to under- or over-representation in sparse or densely detailed areas. The reliance on Farthest Point Sampling for initialization can sometimes result in suboptimal placements in highly irregular geometries. Moreover, our evaluation is currently limited to synthetic datasets, which may not capture real-world variations in lighting, noise, and scale. In particular, preliminary tests on street scene datasets, such as those from KITTI or Cityscapes, reveal significant shortcomings: the method struggles with large-scale unbounded environments, dynamic elements like

Table 1: Comprehensive comparison with 3D Gaussian Splatting on the NeRF synthetic dataset. Higher is better for PSNR and SSIM, lower is better for LPIPS. Best results are shown in bold, second-best are underlined.

Metric	Method	Lego	Chair	Drums	Ficus	Hotdog	Materials	Mic	Ship	Avg
PSNR ↑	3DGS	3.01	2.34	3.22	2.65	3.88	9.19	2.78	4.23	3.91
	Scaffold	35.69	34.99	26.44	35.21	**37.76**	30.65	36.42	31.17	31.83
	2DGS	35.10	**35.05**	26.05	**35.57**	37.36	29.74	35.09	30.60	33.07
	Ours	**35.85**	**35.05**	**26.66**	35.41	37.53	**30.71**	**36.49**	**31.21**	**33.61**
SSIM ↑	3DGS	0.249	0.178	0.122	0.034	0.301	0.167	0.074	0.293	0.177
	Scaffold	0.971	**0.985**	0.812	0.942	**0.983**	0.882	**0.991**	0.891	0.899
	2DGS	0.937	0.861	0.725	0.965	0.972	0.723	0.851	0.711	0.843
	Ours	**0.981**	**0.985**	**0.823**	**0.986**	**0.983**	**0.958**	**0.991**	**0.902**	**0.951**
LPIPS ↓	3DGS	0.512	0.536	0.608	0.781	0.490	0.580	0.661	0.472	0.580
	Scaffold	0.012	0.015	0.023	0.033	0.022	**0.034**	0.008	0.032	0.024
	2DGS	**0.011**	**0.012**	0.032	**0.023**	**0.011**	0.037	0.023	0.042	0.024
	Ours	0.018	**0.012**	**0.021**	**0.023**	0.023	0.042	**0.007**	**0.027**	**0.021**

Table 2: Performance vs. number of control points on selected scenes (PSNR / SSIM).

Scene	256 pts	512 pts	1024 pts
Chair	34.98 / 0.984	35.05 / 0.985	35.06 / 0.985
Materials	30.21 / 0.945	30.71 / 0.958	30.73 / 0.959
Drums	26.18 / 0.809	26.66 / 0.823	26.68 / 0.824

Table 3: Transformer architecture comparison (average on NeRF synthetic).

Config	PSNR ↑	SSIM ↑	Memory
1L-2H	33.42	0.948	3.1GB
2L-4H (Ours)	**33.61**	**0.951**	3.8GB
4L-4H	33.63	0.951	5.2GB
2L-8H	33.64	0.952	4.9GB

Table 4: Loss weight sensitivity (average PSNR on 4 scenes).

λ_1	λ_2	λ_3	PSNR	SSIM
0.01	0.01	0.01	33.21	0.946
0.10	0.05	0.02 (Ours)	**33.61**	**0.951**
0.15	0.03	0.01	33.58	0.950
0.30	0.10	0.05	32.87	0.943

Table 5: Progressive component addition ablation.

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓
Scaffold-GS Baseline	33.12	0.945	0.026
+ Control Points	33.31	0.947	0.024
+ Transformer	33.45	0.949	0.023
+ Control Contrast	33.52	0.950	0.022
+ Gaussian Contrast (Full)	**33.61**	**0.951**	**0.021**

moving vehicles, and varying weather conditions, leading to artifacts in reconstructed scenes and lower metrics. This highlights a lack of generalization to outdoor, real-world scenarios where SfM points are noisier and less dense. Computational costs, while moderate, could increase with more control points or larger transformers, warranting further analysis. Future work could address these by exploring adaptive control point allocation based on scene density, integration with real-world datasets for broader evaluation, and optimizations for efficiency in diverse environments.

7 Conclusion

We presented SCGC, an extension of Scaffold-GS that leverages transformer optimization and contrastive learning to enhance anchor distribution and geometric consistency in 3D scene reconstruction. Our method introduces a transformer-based control point optimization mechanism, a dual-level contrastive learning scheme, and a joint optimization strategy, building on Scaffold-GS's hierarchical anchor system. This approach maintains the baseline's efficiency while adding geometric guidance through sparse control points, resulting in consistent quality improvements, particularly in challenging regions with complex geometry or limited texture. The use of Farthest Point Sampling for control point initialization and k-nearest neighbor assignment to link these points with anchors ensures a structured representation that outperforms Scaffold-GS in synthetic datasets like NeRF, with gains in PSNR and SSIM metrics.

Despite these advances, the method faces limitations that suggest areas for future exploration. The fixed number of control points restricts scalability, and the reliance on synthetic data highlights the need for real-world validation. Preliminary tests on street scenes reveal struggles with unbounded environments and dynamic elements, indicating a gap in generalization that could be addressed with adaptive techniques. The incremental nature of the improvements also suggests that while SCGC enhances structured representations through attention mechanisms, its impact is most pronounced in controlled settings. Future research could focus on adaptive control point allocation to better handle varying scene densities, extend the framework to dynamic scenes with moving objects, and conduct a comprehensive computational cost analysis to optimize performance. Overall, SCGC represents a promising step toward more robust neural rendering, with potential to evolve into a versatile tool for diverse 3D reconstruction tasks.

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